Advanced Empirical Finance: Topics and Data Science

Machine learning foundations

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Today

- What machine learning adds to empirical finance
- The bias-variance trade-off: ridge and lasso
- Cross-validation for time series
- Machine learning workflows in R and Python

What machine learning adds to empirical finance

TBD

The bias-variance trade-off: ridge and lasso

TBD

Cross-validation for time series

TBD

Machine learning workflows in R and Python

TBD

Before the next lecture

- **Exercise 6:** Shrinkage estimation
- **Read:** Kozak, Nagel & Santosh (2020)